

1 Training inputs (same source-keyframe generation)

Source keyframe generation



Null-edited negative control
(label = 0)



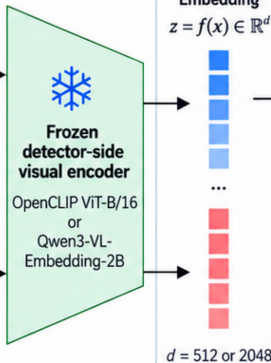
Generator-matched edit
(e.g., null edit)

Trigger-edited positive
(label = 1)



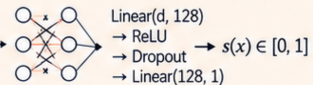
Trigger-guided edit
(e.g., add object)

2 Frozen visual encoder



3 TAM-VLM detection head (trainable)

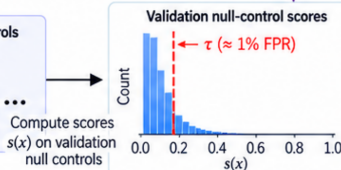
One-hidden-layer MLP (lightweight)



Detector score
 $s(x) \in [0, 1]$

4 Validation and threshold selection

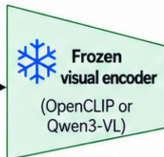
Validation null controls
(label = 0)



Validation-derived low-FPR threshold τ
Set τ using validation null controls (target $\approx 1\%$ FPR)

5 Run-time input screening

Incoming image
(at run time)



Embedding
 $z = f(x) \in \mathbb{R}^d$



Trained
TAM-VLM head

$s(x) \in [0, 1]$

Compare to τ
Alert if
 $s(x) > \tau$

Alert
(flag for external
response policy)

No alert
(forward to unchanged
victim VLM)



Training and validation notes

1. Visual encoder frozen.
2. Only detection head trained.
3. Checkpoint selected by validation AUROC.
4. Threshold selected using validation null controls (target $\approx 1\%$ FPR).
5. No test-set threshold tuning.

Legend



Frozen module
(parameters not updated)



Trainable module
(only detection head)

→ Data flow

--- Used for threshold selection
(validation only)